北京交通大學 BEIJING JIAOTONG UNIVERSITY

信号与系统

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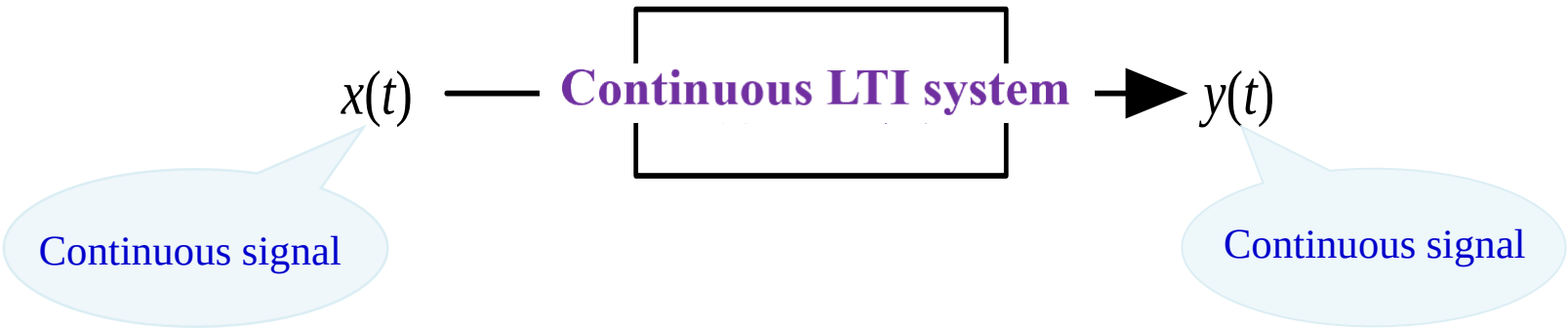
Time domain description of the LTI system

- Time domain description of a continuous LTI system

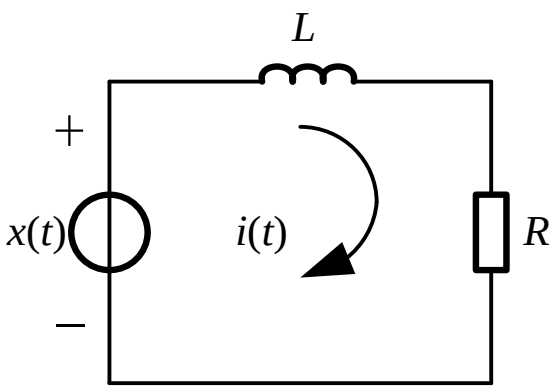
- Time domain description of discrete LTI systems

Time domain of a continuous LTI system

Continuous LTI system



Time domain of a continuous LTI system



LR circuit

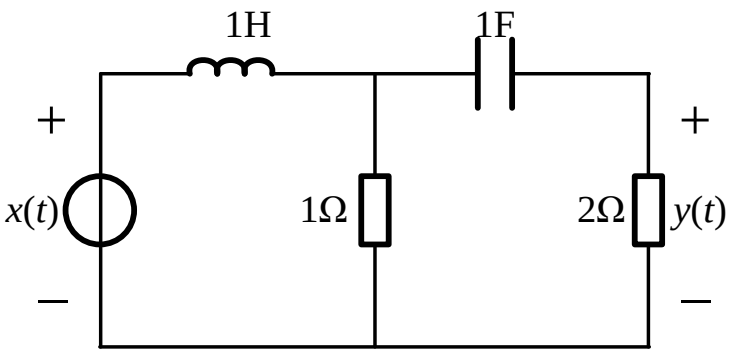
input: $x(t)$; output: $i(t)$

According to KVL,

$$L\frac{\mathrm{d}i(t)}{\mathrm{d}t}+Ri(t)=x(t)$$

The input-output relationship of an LR first-order circuit system can be described by first-order differential equations

Time domain of a continuous LTI system



LRC circuit

input: $x(t)$; output: $y(t)$

According to the basic theory of circuits

$$3\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2}+3\frac{\mathrm{d}y(t)}{\mathrm{d}t}+y(t)=2\frac{\mathrm{d}x(t)}{\mathrm{d}t}$$

The input-output relationship of an LRC second-order circuit system, described by second-order differential equations

Time domain of a continuous LTI system

Mathematical models describing different continuous-time systems are similar, their input-output relationships can be described using differential equations.

The input-output relationship of a continuous-time LTI system can be described by a differential equation with linear constant coefficients

$$y^{(n)}(t)+a_{n-1}y^{(n-1)}(t)+\cdots+a_1y'(t)+a_0y(t)$$
$$=b_mx^{(m)}(t)+b_{m-1}x^{(m-1)}(t)+\cdots+b_1x'(t)+b_0x(t)$$

Time domain of a continuous LTI system

Continuous LTI systems have linear and non-time-varying characteristics and therefore have:

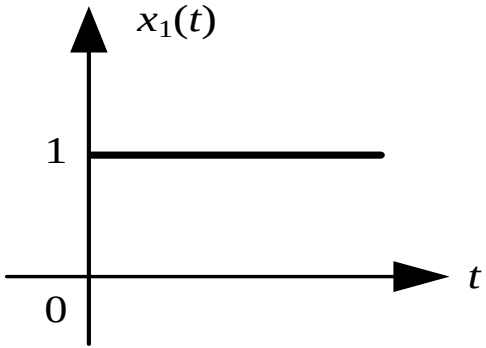
※ Differential properties

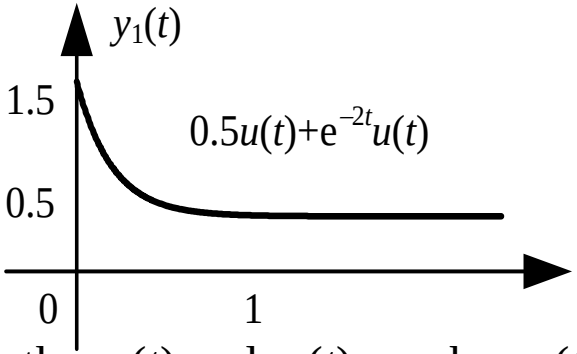
If $x(t) \longrightarrow y(t)$ then $\dfrac{\mathrm{d}x(t)}{\mathrm{d}t} \longrightarrow \dfrac{\mathrm{d}y(t)}{\mathrm{d}t}$

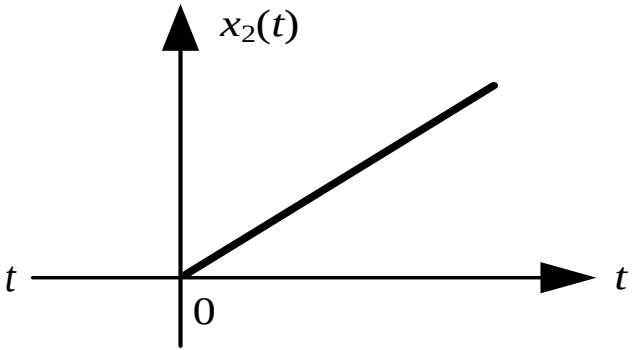
※ Integral characteristics

If $x(t) \longrightarrow y(t)$ then $\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \longrightarrow \int_{-\infty}^{t} y(\tau)\mathrm{d}\tau$

[Example] It is known that a continuous LTI system produces a response of $y_1(t)$ under $x_1(t)$ excitation, try to find the response $y_2(t)$ produced by the system under the $x_2(t)$ stimulus.







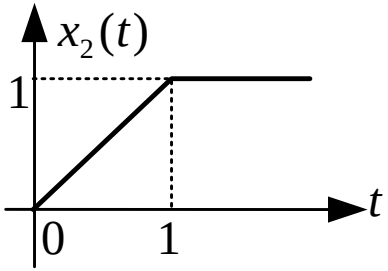
As can be seen from the $x_1(t)$ and $x_2(t)$ graphs, $x_2(t)$ has the following relationship with $x_1(t)$

$$x_2(t) = x_1^{(-1)}(t) = \int_{-\infty}^{t} x_1(\tau)\mathrm{d}\tau$$

Therefore, the same relationship exists between $y_1(t)$ and $y_2(t)$.

$$y_2(t) = y_1^{(-1)}(t) = \int_{-\infty}^{t} y_1(\tau)\mathrm{d}\tau = 0.5tu(t) + 0.5(1-\mathrm{e}^{-2t})u(t)$$

[Example] Signal $x_1(t)=u(t)-u(t-1)$ is known to respond to zero-state through a continuous LTI system $y_1(t)=\delta(t-1)-\delta(t-2)$. Try to generate the response $y_2(t)$ of the system under $x_2(t)$ excitation.



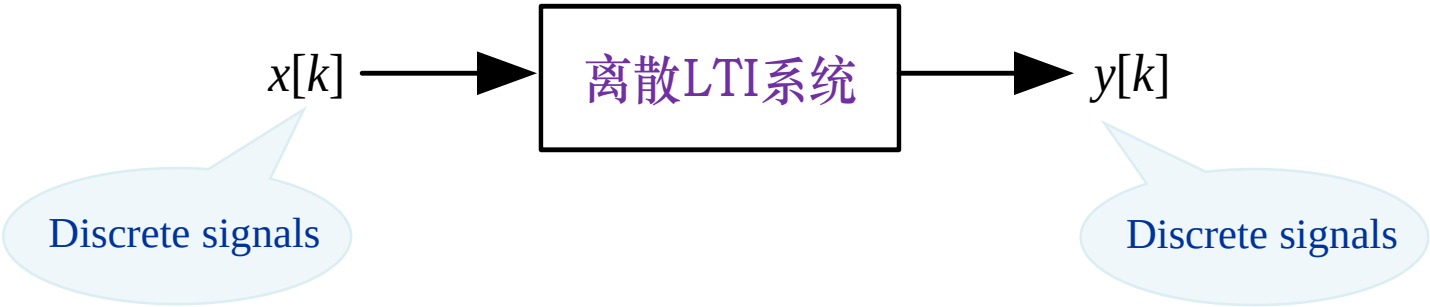
解:

$$x_2(t)=x_1^{(-1)}(t)=\int_{-\infty}^{t} x_1(\tau)\mathrm{d}\tau$$

$$y_2(t)=y_1^{(-1)}(t)=\int_{-\infty}^{t} y_1(\tau)\mathrm{d}\tau=u(t-1)-u(t-2)$$

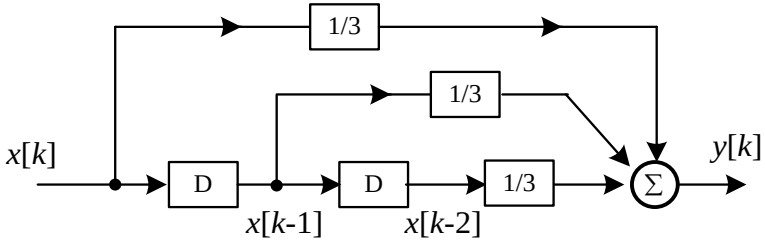
Time domain of discrete LTI systems

Discrete LTI systems



Time domain of discrete LTI systems

Moving average system

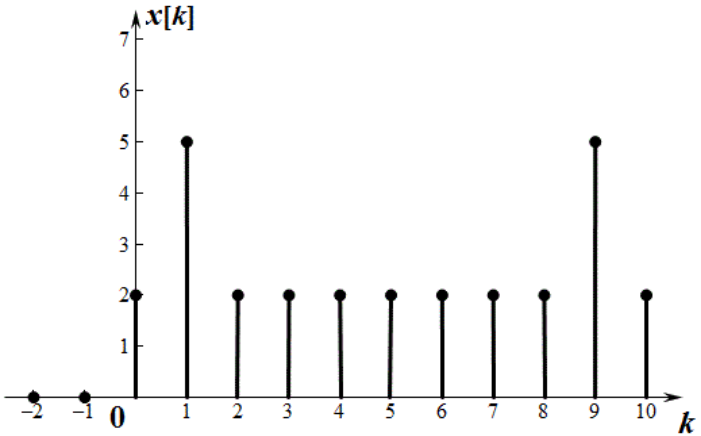


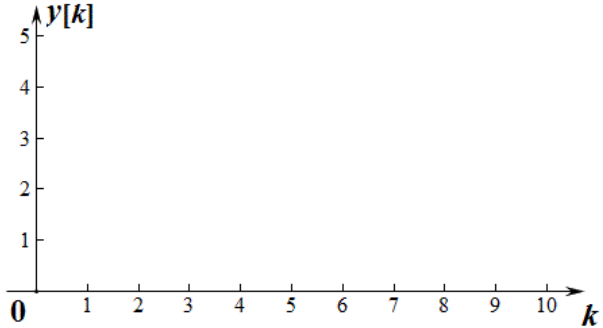
3-point moving average system

$$y[k]=\frac{1}{3}\{x[k]+x[k-1]+x[k-2]\}$$

$$(x[-2]=0\ ,\ \ x[-1]=0)$$

The input-output relationship of a discrete system can be described by the difference equation





Time domain of discrete LTI systems

The mathematical models of different discrete-time systems are similar and can all be described by difference equations.

Discrete systems can be described using forward and backward difference equations. If the input/output signal is causal, the backward difference equation is generally used

Discrete LTI systems are generally described using the linear constant coefficient difference equation

$$a_0 y[k] + a_1 y[k-1] + \cdots + a_{n-1} y[k-n+1] + a_n y[k-n]$$
$$= b_0 x[k] + b_1 x[k-1] + \cdots + b_{m-1} x[k-m+1] + b_m x[k-m]$$

Time domain of discrete LTI systems

Discrete LTI systems are generally described using the linear constant coefficient difference equation

$$a_0 y[k] + a_1 y[k-1] + \cdots + a_{n-1} y[k-n+1] + a_n y[k-n]$$
$$= b_0 x[k] + b_1 x[k-1] + \cdots + b_{m-1} x[k-m+1] + b_m x[k-m]$$

or abbreviated $$\sum_{i=0}^{n} a_i y[k-i] = \sum_{j=0}^{m} b_j x[k-j]$$

$a_i\,(i = 0,1,\cdots n), b_j\,(j = 0,1,\cdots,m)$ is the coefficients, a_0=1

Example: A moving average system using M points filters out signal noise.

Solution: The input-output relationship of the moving average system of M points is

$$y[k]=\frac{1}{M}\sum_{n=0}^{M-1} x[k-n]$$

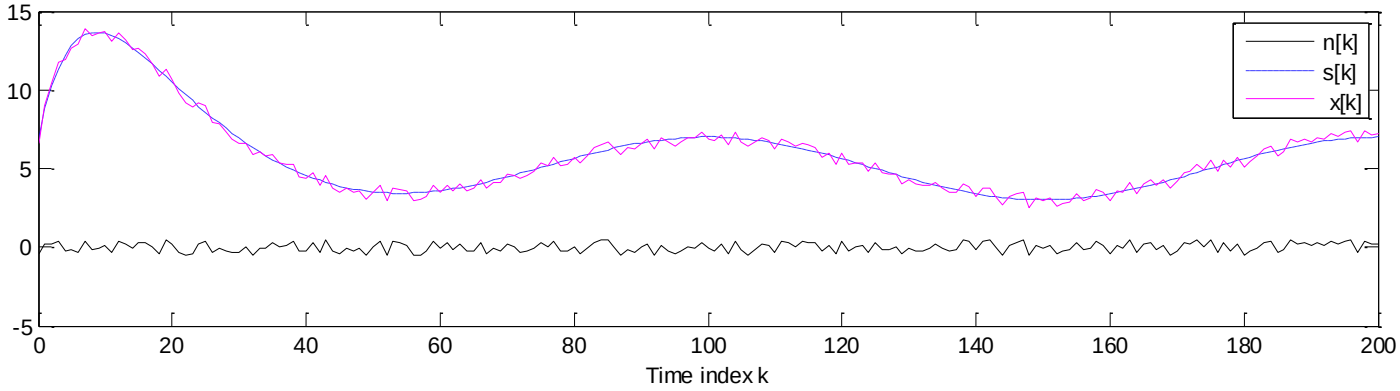
Raw signal: $s[k]=5+2\cos(0.02\pi k)+(2k)0.9^k$

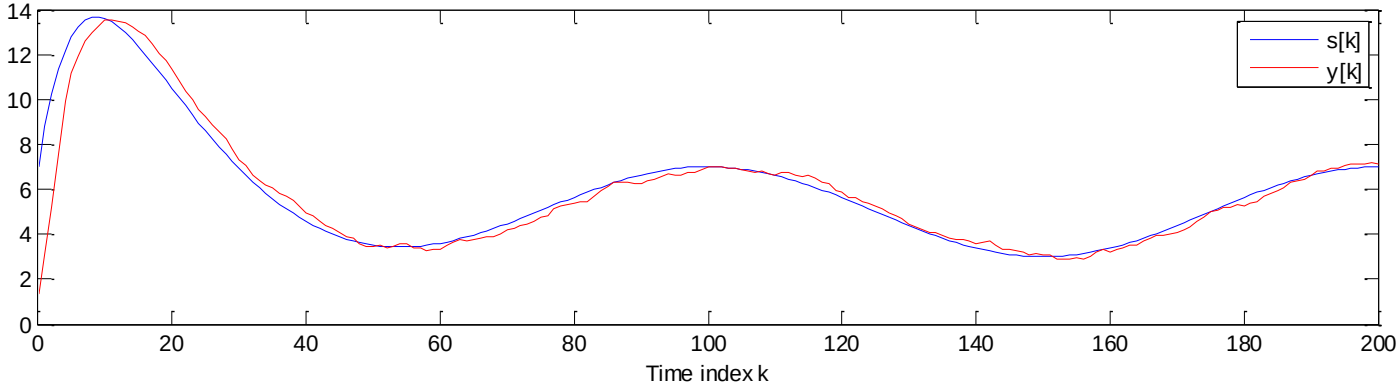
Noisy signal: $n[k]$

Noise-cancel signal: $x[k]=s[k]+n[k]$

A moving average system using the M point filters out noisy signal n[k] from signal x[k]

Example: A moving average system using M points filters out signal noise.





Example: A moving average system using M points filters out signal noise.

```
% Signal Smoothing by Moving Average Filter
N = 201;
n = 1.0*rand(1,N)-0.5;
k=0:N-1;
s=2*k.*(0.9.^k)+2.0*cos(0.02*pi*k)+5.0;
x=s+n;
subplot(2,1,1);
plot(k,n,'k-', k,s,'b--', k,x,'m-');
xlabel('Time index k');  legend('n[k]','s[k]', ' x[k]');
M =5;  b = ones(M,1)/M;   a =[1];
y = filter(b,a,x);
subplot(2,1,2);
plot(k,s,'b-', k,y,'r-');
xlabel('Time index k');  legend('s[k]','y[k]');
```

Time domain of discrete LTI systems

Discrete LTI systems have linear and non-temporal characteristics

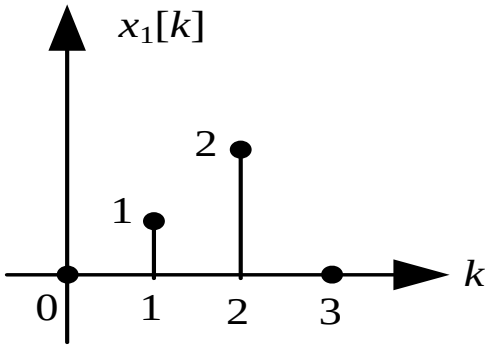
※ Differential characteristics

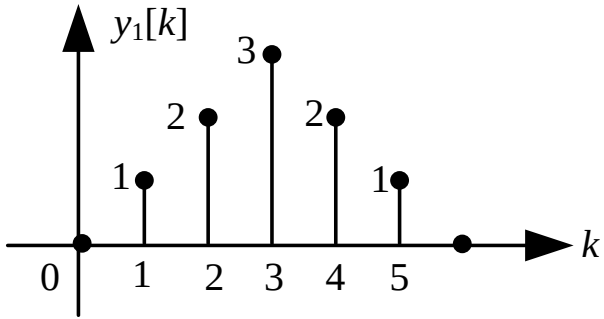
If $x[k] \longrightarrow y[k]$ then $\nabla x[k] \longrightarrow \nabla y[k]$

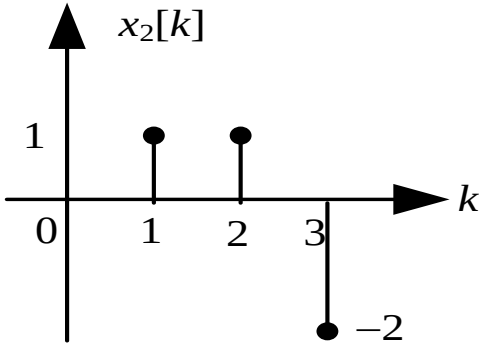
※ Summation characteristics

If $x[k] \longrightarrow y[k]$ then $\sum_{n=-\infty}^{k} x[n] \longrightarrow \sum_{n=-\infty}^{k} y[n]$

[Example] It is known that the discrete-time LTI system produces a response of $y_1[k]$ at $x_1[k]$ excitation. Try to find the response $y_2[k]$ generated by the system under $x_2[k]$ excitation.







As can be seen from the $x_1[k]$ and $x_2[k]$ graphs, $x_2[k]$ has the following relationship with $x_1[k]$.

$$x_2[k] = \nabla x_1[k] = x_1[k] - x_1[k-1]$$

Therefore, the same relationship exists between $y_2[k]$ and $y_1[k]$

$$y_2[k] = \nabla y_1[k] = y_1[k] - y_1[k-1] = \{1,1,1,-1,-1,-1; k = 1,2,3,4,5,6\}$$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!